Solving Quadratics by Factoring

Algebra I | Guided Notes — we fill these in together

Name: ____________________________ Date: ________________

Vocabulary — write the definition

Zero Product Property:

____

____

Root (solution):

____

____

We do together — worked example

Solve: $x^2 + 5x + 6 = 0$

Step 1. Get one side equal to 0. (already done)

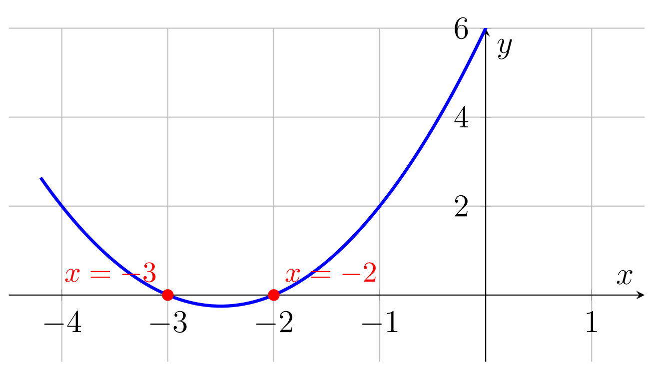
Step 2. Factor the left side. $x^2 + 5x + 6 = (\_\_\_\_\_\_\_\_)(\_\_\_\_\_\_\_\_)$

Step 3. Zero Product Property — set *each* factor equal to 0.

$\_\_\_\_\_\_\_\_\_\_ = 0$ $\_\_\_\_\_\_\_\_\_\_ = 0$

Step 4. Solve each little equation. $x = \_\_\_\_\_\_$ $x = \_\_\_\_\_\_$

See it on the graph. The solutions are exactly where the parabola crosses the x-axis.



The graph crosses at $x = -2$ and $x = -3$ — the same answers you found in Step 4.

You Try. Solve $x^2 - x - 12 = 0$. Show every step, then name the two roots.